FINITENESS OF ETALE FIBRES OVER A POINT: ONE COVERING FAMILY VERSUS EVERY AFFINE ETALE MORPHISM

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Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Etale morphisms are open. Remark remark-recall (same chapter, structural). The etale maps q_i are open, so their images cover U.
- Bounded fibres over quasi-compact source. Morphisms, lemma locally-quasi-finite-qc-source-universally-bounded (cross-chapter, named). Fibres of p_i over the quasi-compact opens are finite.
- Fibres of the presentation. Properties of Spaces, lemma points-cartesian (cross-chapter, named). Relates fibres of phi over x to the pieces over the U_i.

Source reference. Original reference 03JA: focused-seeds/geometry/S9-decent-u-finite-above-x/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-u-finite-above-x/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $x \in |X|$. The following are equivalent:*

1. *there exists a family of schemes U_i and étale morphisms $\varphi_i : U_i \to X$ such that $\coprod \varphi_i : \coprod U_i \to X$ is surjective, and such that for each i the fibre of $|U_i| \to |X|$ over x is finite, and*
2. *for every affine scheme U and étale morphism $\varphi : U \to X$ the fibre of $|U| \to |X|$ over x is finite.*

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Proof. The implication (2) $\Rightarrow$ (1) is trivial. Let $\varphi_i : U_i \to X$ be a family of étale morphisms as in (1). Let $\varphi : U \to X$ be an étale morphism from an affine scheme towards X. Consider the fibre product diagrams

$$\begin{array}{ccc} U \times_X U_i & \xrightarrow[p_i]{} & U_i \\ {\scriptstyle q_i}\downarrow & & \downarrow{\scriptstyle \varphi_i} \\ U & \xrightarrow{\varphi} & X \end{array} \qquad \begin{array}{ccc} \coprod U \times_X U_i & \xrightarrow[\coprod p_i]{} & \coprod U_i \\ {\scriptstyle \coprod q_i}\downarrow & & \downarrow{\scriptstyle \coprod \varphi_i} \\ U & \xrightarrow{\varphi} & X \end{array}$$

Since q_i is étale it is open (see Remark 03II). Moreover, the morphism $\coprod q_i$ is surjective. Hence there exist finitely many indices $i_1, \ldots, i_n$ and a quasi-compact opens $W_{i_j} \subset U \times_X U_{i_j}$ which surject onto U. The morphism p_i is étale, hence locally quasi-finite (see remark on étale morphisms above). Thus we may apply Morphisms, Lemma 03JA to see the fibres of $p_{i_j}|_{W_{i_j}} : W_{i_j} \to U_i$ are finite. Hence by Properties of Spaces, Lemma 03H4 and the assumption on φ_i we conclude that the fibre of φ over x is finite. In other words (2) holds. $\square$